Introduction

This test program intends to demonstrate the seal performance under site conditions after several reports of failure and inconsistency of leakage.

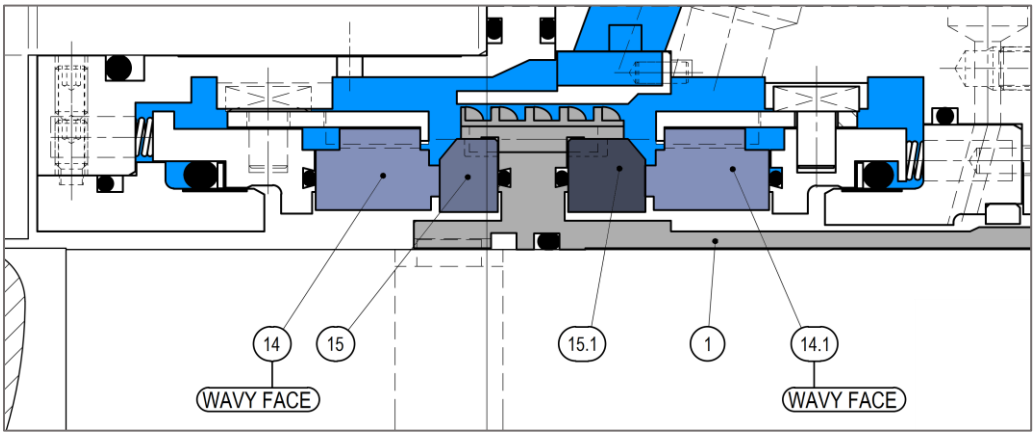
The subject of this testing is mechanical seal order 950969 modified. This version of mechanical seal was designed based on new seal faces geometry to include lasered wavy face technology.

Test N°1 – (Aug 23 – Aug 29, 2023)

First test intends to confirm calculation results based on same operating conditions on site. Although, barrier fluid has been changed to Mono-Ethylene Glycol + water mixture, this test will be conducted with water, change in barrier fluid will be introduced in next test run.

Seal Configuration

Seal faces			
	Part Drawing	Material	Face condition
Sleeve (1)	C0619166NL	Alloy C-276	New
Inner Seal Stator (14)	C0618722SL	Silicon Carbide (SL)	New
Inner Seal Rotor (15)	C0619339SL	Silicon Carbide (SL)	New
Outer Seal Stator (14.1)	C0618722SL	Silicon Carbide (SL)	New
Outer Seal Rotor (15.1)	C0618723SL	Silicon Carbide (SL)	New



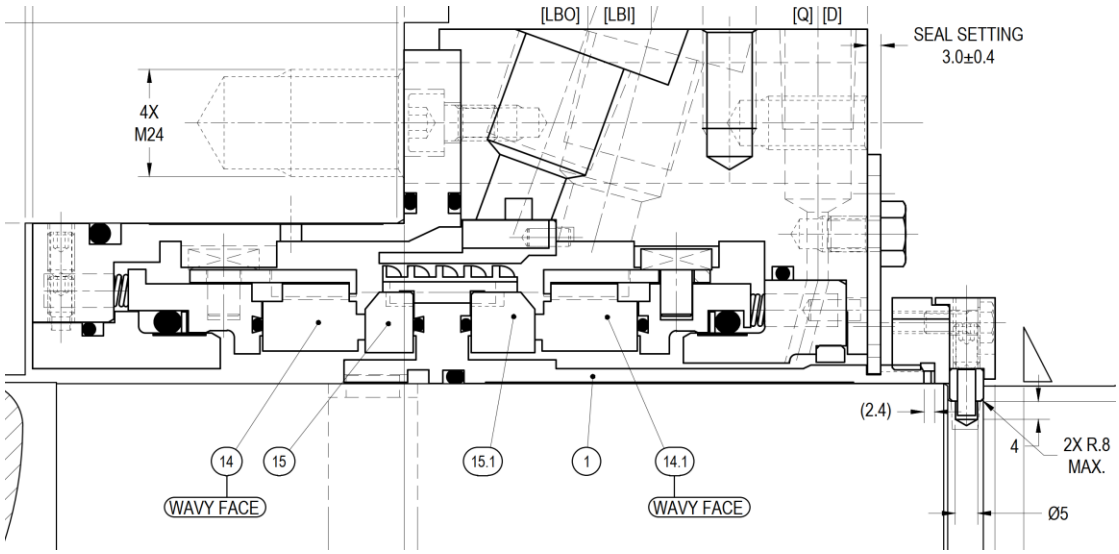
Cross Section of The New Parts

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Test Conditions & Results

	Job seal barrier
Duration	Minimum 72h
Speed	2950 rpm
Fluid	Water
Pressure	70 barg
Inlet temp.	40 °C
Outlet temp.	43 °C
Inner Seal Leakage	Avg. 16.1 ml/h
Outer Seal Leakage	Avg. 14.4 ml/h
Flow	13.2-12.2 l/min

Tester Drawing D0571813

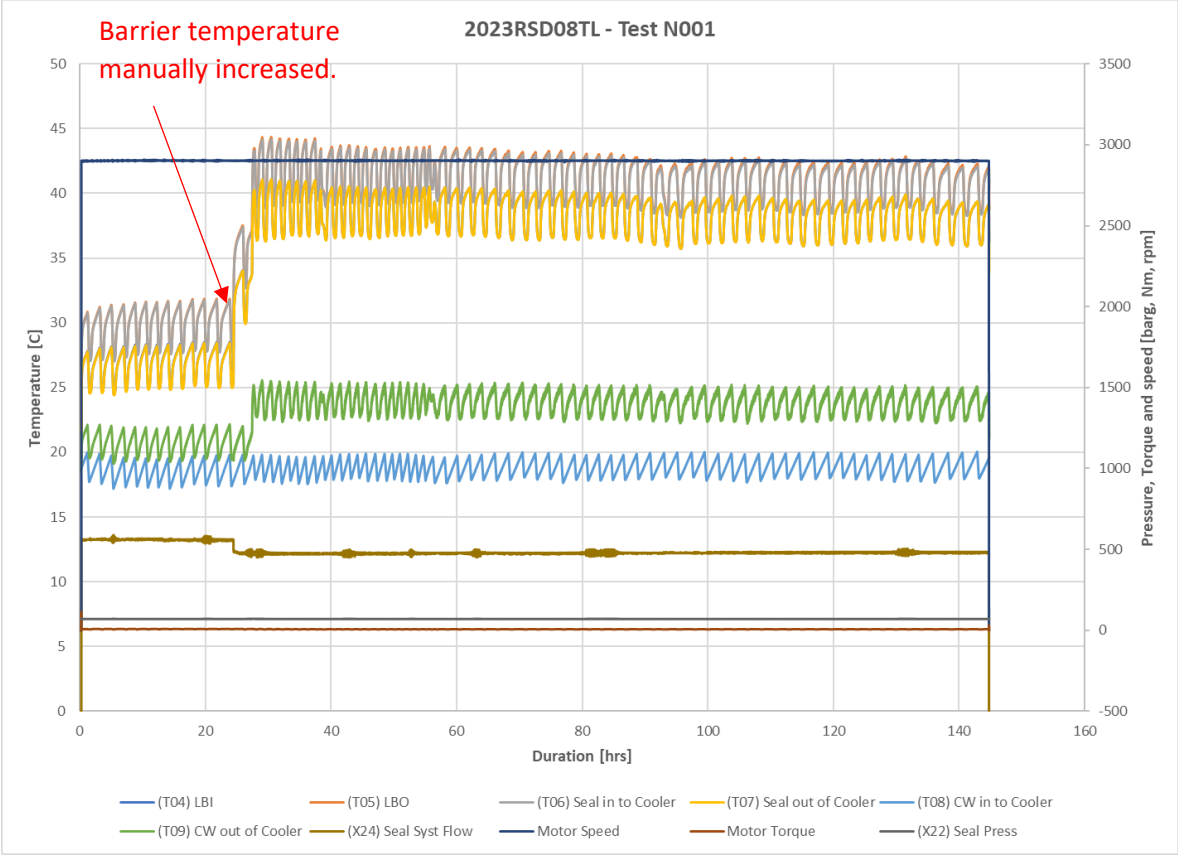


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Tester unit B

Logged Data

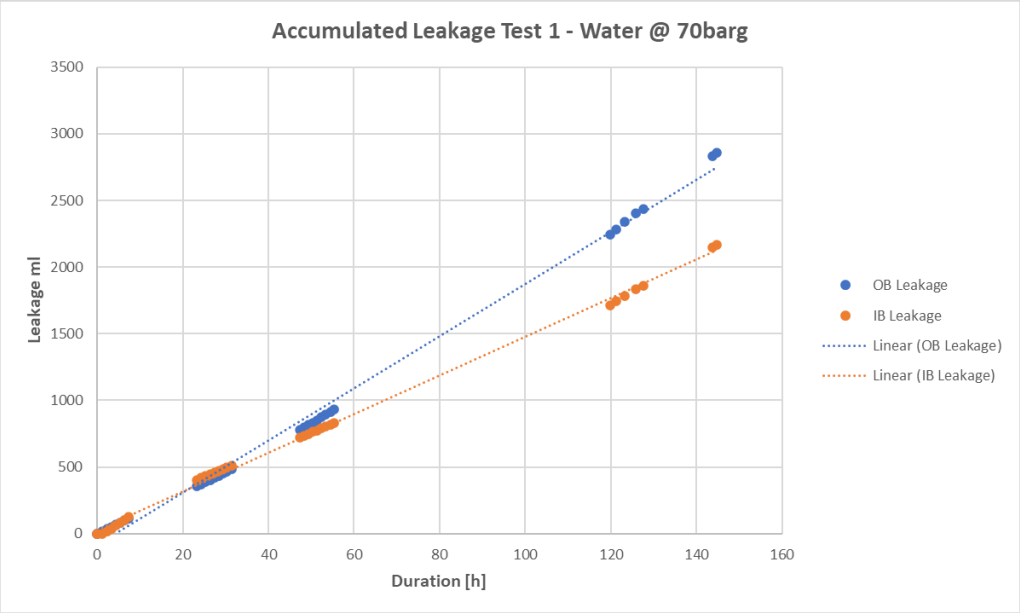


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The first 24h showed stable performance, however, barrier temperature was too low reason for reducing the cooling water flow to have higher barrier temperature.

Leakage was steady during the whole testing period.



Manually Logged Data

Test Project Number:	2023RSD08TL	Run no.:	TCN001	OB		IB	
Test Unit:	B	ire Rate s/S on 60 trigger on 5		Seal Type:	UHTW/UHTW	Seal Type:	UHTW/UHTW
Speed:	CCW	Aligned	yes	Stationary Face:	C618722SL	Stationary Face:	C618722SL
Pres / Temp:		Cooling CTRL Valve 0% is complete open		Rotating Face:	C0619323SL	Rotating Face:	C0619339SL
Barrier Medium:	Demi water	Flow CTRL Valve 100% is complete open					

											OB				IB			
Date	Time	Running Hours Total	Runned Hours after a start	By	RPM	Torque [Nm]	Syst Flow [l/min]	Seal Press [barg]	LBI Temp. [°C]	LBO Temp. [°C]	OB total leakage in the glasss	OB Leakage [ml]	Over [hours]	OB leakage Rate [ml/hr]	IB total leakage in the glass	IB Leakage [ml]	Over [hours]	IB Leakage Rate [ml/hr]
23-Aug-23	8:09	0.00	0.00	Rob & Claude	0	0.00	0.00	70.00	21.90	21.96	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
23-Aug-23	8:24	0.00	0.00	Rob & Claude	2900	7.60	13.33	70.00	22.72	25.44	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
23-Aug-23	9:36	1.20	1.20	Rob & Claude	2900	5.70	13.26	69.94	24.64	27.14	0.00	15.00	1.20	12.50	0.00	0.00	1.20	0.00
23-Aug-23	10:34	2.42	2.42	Rob & Claude	2900	7.68	13.25	70.10	27.21	30.34	0.00	18.00	1.22	14.79	0.00	19.50	1.22	16.03
23-Aug-23	11:29	3.33	3.33	Rob & Claude	2900	5.89	13.32	70.06	25.03	27.64	0.00	14.50	0.92	15.82	0.00	17.00	0.92	18.55
23-Aug-23	12:25	4.27	4.27	Rob & Claude	2900	7.90	13.25	69.98	27.25	30.45	0.00	17.80	0.93	19.07	0.00	23.50	0.93	25.18
23-Aug-23	13:28	5.32	5.32	Rob & Claude	2900	6.00	13.35	80.50	24.50	27.10	0.00	15.50	1.05	14.76	0.00	21.00	1.05	20.00
23-Aug-23	14:28	6.32	6.32	Rob & Claude	2900	8.00	13.26	69.90	27.47	30.75	0.00	17.00	1.00	17.00	0.00	21.00	1.00	21.00
23-Aug-23	15:31	7.37	7.37	Rob & Claude	2900	6.20	13.26	69.87	24.90	27.53	0.00	17.50	1.05	16.67	0.00	20.80	1.05	19.81
24-Aug-23	7:30	23.35	23.35	Rob & Claude	2900	8.22	13.27	70.00	28.11	31.46	0.00	240.00	15.98	15.02	0.00	279.00	15.98	17.46
24-Aug-23	**8:30**	24.35	24.35	Rob & Claude	2900	6.64	13.25	70.20	25.28	28.07	0.00	10.50	1.00	10.50	0.00	16.75	1.00	16.75
24-Aug-23	9:30	25.35	25.35	Rob & Claude	2900	7.84	12.21	70.00	33.49	36.97	0.00	21.00	1.00	21.00	0.00	13.00	1.00	13.00
24-Aug-23	10:31	26.37	26.37	Rob & Claude	2900	6.12	12.24	70.00	30.05	32.87	0.00	15.00	1.02	14.75	0.00	15.00	1.02	14.75
24-Aug-23	**11:33**	27.40	27.40	Rob & Claude	2900	7.83	12.12	70.60	33.79	37.24	0.00	16.80	1.03	16.26	0.00	13.20	1.03	12.77
24-Aug-23	12:31	28.37	28.37	Rob & Claude	2900	7.47	12.25	70.50	38.19	41.34	0.00	17.00	0.97	17.59	0.00	13.80	0.97	14.28
24-Aug-23	13:30	29.35	29.35	Rob & Claude	2900	5.62	12.25	70.00	36.46	39.13	0.00	15.80	0.98	16.07	0.00	10.50	0.98	10.68
24-Aug-23	14:25	30.27	30.27	Rob & Claude	2900	7.34	12.19	70.20	40.95	44.23	0.00	16.20	0.92	17.67	0.00	12.00	0.92	13.09
24-Aug-23	15:32	31.38	31.38	Rob & Claude	2900	7.47	12.22	70.10	40.54	43.80	0.00	18.50	1.12	16.57	0.00	13.75	1.12	12.31
25-Aug-23	7:30	47.35	47.35	Rob & Claude	2900	5.88	12.13	70.00	36.68	39.36	0.00	290.00	15.97	18.16	0.00	210.00	15.97	13.15

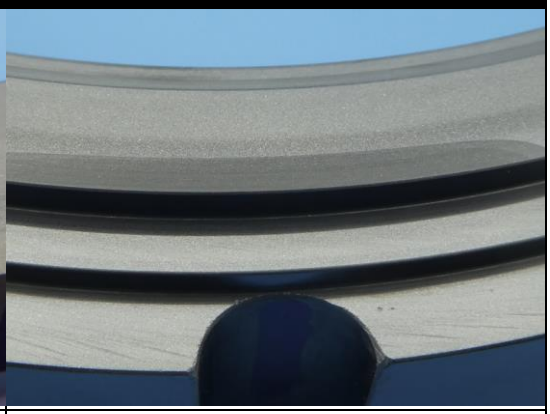
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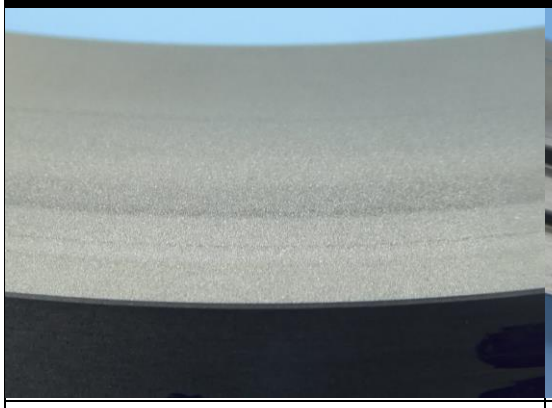
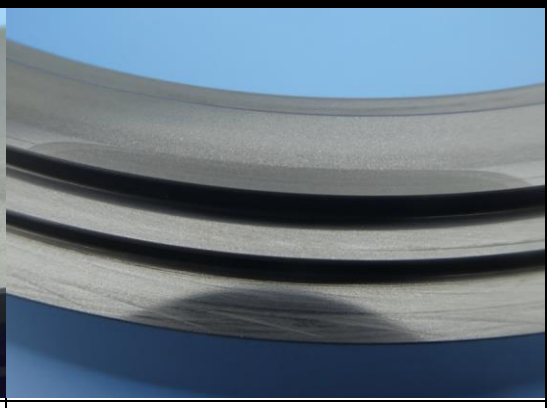
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Measurements Results

Leakage rate		
	Inner Seal	Outer Seal
Total Run time [h]	144.6	144.6
Total Seal Leakage [ml]	2857.6	2164.3
Seal Leakage Rate [ml/h]	19.7	14.9

Post-Test Pictures

IB Rotor	IB Stator
	
Ra	Ra

OB Rotor	OB Stator
	
Ra	Ra

Test Conclusion

The first test with newly designed faces (wavy faces) on water has been very successful, the performance in terms of leakage stability and temperature was acceptable.

Seal faces conditions after the test didn't show any damage or deterioration from running over 144 hours on water. It is therefore considered an acceptable seal faces combination.

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